RACE
Regular Analysis Through Continuous Exercise

Chemical Bonding : RACE-01

Kossel -Lewis approach to Chemical Bonding; Octet Rule, Lewis Dot Structure

1. Pick out among the following species isoelectronic with CO_2.

(A) $NO_3^{\ominus}$ (B) $(CNO)^-$ (C) $(NCN)^{2-}$ (D) All of these

CCB436

2. Match the column :

Column I	**Column II**
(A) $NH_3.BF_3$	(P) Ionic bond
(B) CO	(Q) Covalent bond
(C) NH_4Cl	(R) Co-ordinate bond
(D) KI_3	(S) 3 lone pair on any one atom

CCB443

Limitations of Octet Rule (Hypervalent, Hypovalent, Odd Electron Species)

3. How many compounds violate octet rule ?

(i) CO_2 (ii) PCl_5 (iii) SiF_4 (iv) BrF_5
(v) IF_7 (vi) PCl_3 (vii) H_2SO_4 (viii) BF_3

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CCB549

Types of Covalent Bond and Overlapping of Orbitals; Orbital Overlap along Different Axes

4. If y-axis is the approaching axis between two atoms, then which of the set of orbitals cannot form the π bond between two atoms in general.

(A) $p_z - p_z$ (B) $p_x - p_x$ (C) $p_x - p_y$ (D) None of these

CCB435

5. Which of the following set of overlap cannot provide π-bond formation.

(A) 3d and 2p (B) 2p and 3p (C) 2p and 2p (D) 3p and 1s

CCB437

6. Indicate the wrong statement according to Valence bond theory:

(A) A sigma bond is stronger then π-bond
(B) p-orbitals always have only sidewise overlapping
(C) s-orbitals never form π-bonds
(D) There can be only one sigma bond between two atoms

CCB438

7. If the molecular axis is z then which of the following overlapping is not possible.

(A) $p_z + p_z$ = σ bond (B) $p_x + p_y$ = π bond (C) $p_x + p_x$ = π bond (D) $p_y + p_y$ = π bond

CCB439

8. If molecular axis is X then which of the following overlapping will form π bond.

$p_z + p_z$, $p_x + p_x$, $p_x + p_y$, $s + p_z$, $p_y + p_y$

CCB444

For More Join: @IITJEE_Advanced

9. If X-axis is the molecular axis, then p-molecular orbitals are formed by the overlap of

(A) $s + p_z$ (B) $p_x + p_y$ (C) $p_z + p_z$ (D) $p_x + p_x$

CCB480

Order of Relative Strength of Covalent Bonds (σ and π bond)

Paragraph for question nos. 10 to 11

Different types of bonds are formed in the chemical compounds. These bond have different strength and bond energies associated with them. These bonds are formed with atoms in different environments.

10. Which of the following bond has highest bond energy?

(A) σ -bond (B) π-bond (C) Hydrogen bond (D) None of these

CCB440

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	(A) Q,R,S ; (B) Q,R ; (C) P,Q,R ; (D) P,Q,R,S	5	C	D	B	B	2	C	A

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Chemical Bonding : RACE-02

Atomic Orbitals Involved in Hybridisation

1. Match the column:

Column -I (Type of orbital)	**Column-II (Orbitals involved in hybridisation)**
(A) d_{z^2} -orbital	(P) sp^3 (Tetrahedral)
(B) s - orbital	(Q) sp^3d^2 (Octahedral)
(C) $d_{x^2-y^2}$ - orbital	(R) sp^3d (TBP)
(D) p_y - orbital	(S) dsp^2 (square planar)

CCB450

2. How many of the following species have no d-orbitals involved in their hybridisation

XeF_2, SnF_2, $PbCl_2$, SF_4, CCl_4, BF_3, SO_2Cl_2, XeO_2F_2, $XeOF_4$, $POCl_3$

CCB454

Solid State Hybridisation

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3. What is hybridisation of central atom of anionic part of PBr_5 in crystalline state.

(A) sp^2 (B) sp^3 (C) sp (D) not applicable

CCB455

4. All possible bond angles in anionic part of PCl_5 are.

(A) 109° 28' only (B) 90°, 180°

(C) 90°, 120°, 180° (D) 72°, 90°, 180°

CCB456

VSEPR Table

5. The compound MX_4 is tetrahedral. The number of < XMX angles in the compound is

(A) Three (B) Four (C) Five (D) Six

CCB445

6. **Statement-1** : CH_4 and CH_2F_2 are having regular tetrahedron geometry.

Statement-2 : Both are having same hybridization.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CCB446

For More Join: @IITJEE_Advanced

7. Which of following pair of species is having different hybridisation but same shape.

(A) $BeCl_2$ and CO_2 (B) CO_2 and SO_2

(C) SO_2 and I_3^+ (D) ICl_2^- and BeH_2

CCB447

Paragraph for Q.8 to 9

According VSEPR theory, in the electronic geometry of the molecule(s), electron pair tend to minimise repulsion and follow the repulsion order in presence of lone pair of electrons with bond pair of electrons.

lp – lp > lp – bp > bp – bp

and similarly, double bond pair of electrons follow the repulsion order with single bond pair of electrons.

double bond – double bond > double bond – single bond > single bond – single bond

8. Which of the following molecular geometry is distorted geometry from their ideal geometry.

(A) PCl_3F_2 (B) SOF_4 (C) $XeF_5^{\ominus}$ (D) XeO_3F_2

CCB451

9. Which of the following statement is **CORRECT** about XeO_2F_2?

(A) XeO_2F_2 is sp^3d hybridized and has lone pair of electrons at axial position of its electronic geometry.

(B) XeO_2F_2 has maximum 10- lone pair of electrons.

(C) XeO_2F_2 has distorted see-saw shape.

(D) XeO_2F_2 contains $d\pi$–$d\pi$ bond.

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CCB452

10. **Statement-1 :** ClF_2^- is linear while ClF_2^+ is bent.

Statement-2 : Cl - atom in ClF_2^- and in ClF_2^+, is having same state of hybridisation.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CCB453

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	(A) Q,R ; (B) P,Q,R,S ; (C) Q,S ;(D) P,Q,R,S	6	D	B	D	D	CD	B	C	C

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Chemical Bonding : RACE-03

Bent's Rule

1. Which of the bond length order data is **INCORRECT** -

(A) P–Cl > P–F in PCl_3F_2 (B) S–$F_{(axial)}$ > S – $F_{(eq)}$ in SF_6

(C) S–$F_{(axial)}$ > S – $F_{(eq)}$ in SF_4 (D) NO_2^- < NO_3^- (N – O)

CCB458

2. O_2F_2 is an unstable yellow orange solid and H_2O_2 is a colourless liquid, both have O–O bond. O–O bond length in H_2O_2 & O_2F_2 is respectively.

(A) 1.22Å, 1.48Å (B) 1.48Å, 1.22Å (C) 1.22Å, 1.22Å (D) 1.48Å, 1.48Å

CCB459

3. Comment on the C – C bond length for C_2H_6 and C_2F_6 compounds:

(A) $d_{C-C}(C_2H_6) > d_{C-C}(C_2F_6)$ (B) $d_{C-C}(C_2F_6) > d_{C-C}(C_2H_6)$

(C) $d_{C-C}(C_2F_6) = d_{C-C}(C_2H_6)$ (D) Can't be predicted

CCB461

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4. Choose the **CORRECT** statements :-

(A) d_{P-F}(axial) > d_{P-F}(equatorial) in PF_3Cl_2

(B) $d_{P-F} > d_{P-Cl}$ in PF_2Cl_3

(C) $d_{P-Cl}(PF_2Cl_3) > d_{P-Cl}(PF_3Cl_2)$

(D) All d_{P-Cl} (in PF_2Cl_3) are identical while all d_{P-F}(in PF_3Cl_2) are not identical

CCB463

5. Identify the pair in which the specified bond length of first is greater than second.

PCl_3F_2, PF_3Cl_2 : $BL_{P-Cl_{eq}}$

SO_2Cl_2, SO_2F_2 : BL_{S-O}

BF_3, BCl_3 : BL_{B-X} X = F/Cl

NO_3^- , NO_2^- : BL_{N-O}

O_3, O_2 : BL_{O-O}

CO, CO_2 : BL_{C-O}

CCB464

Drago's Rule

6. In Which of the following hydrides bond angle is minimum?

(A) H_2O (B) H_2Te (C) H_2S (D) H_2Se

CCB550

For More Join: @IITJEE_Advanced

Odd Electron Molecules and their characteristics

7. Which of the following species has sp^3 hybridisation:-

(A) $\dot{C}H_3$ (B) $\dot{C}F_3$ (C) ClO_2 (D) None of these

CCB469

8. Select the **CORRECT** statement about ClO_3 :

(A) ClO_3 has unpaired electron

(B) ClO_3 has pyramidal structure

(C) In ClO_3, Cl atom is sp^3 hybridised

(D) In ClO_3, Cl atom is sp^2 hybridised

CCB471

9. Which of the following property is/are changed when of (odd electron species) undergoes dimerisation?

(A) Magnetic nature (B) Planarity

(C) Polarity (D) O–F bond length

CCB472

10. Which of the following molecule (s) has/have co-ordinate bond in it's structure?

(A) NO_2 (B) N_2O_5 (C) N_2O_4 (D) N_2O_3 (symm.)

CCB473

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	B	A	ACD	4	B	B	ABC	ABD	ABC

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Chemical Bonding : RACE-04

Molecules which Do Not Exist

1. Which of the following species does not exist?

(A) BF_5^{2-} (B) XeF_4 (C) XeF_5^- (D) XeF_6

CCB457

Dipole Moment; Calculation of Percent Ionic Character

2. Dipole moment of LiF was experimentally determined and was found to be 6.32 D. Calculate percentage ionic character in LiF molecule Li—F bond length is 156 pm.

CCB552

Polarity of Species

Match the column:

3.

Column I	**Column II**
(A) $\dot{Cl}O_2$	(P) Non planar
(B) $\dot{Cl}O_3$	(Q) $\mu \neq 0$
(C) $\dot{N}O_2$	(R) Linear
(D) $\dot{N}O$	(S) planar
	(T) sp^3 hybridisation

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CCB474

4. Of the following molecules, the one, which has permanent dipole moment, is :

(A) SiF_4 (B) BF_3 (C) PF_3 (D) PF_5

CCB553

5. The dipole moment of chlorobenzene is 1.73 D. The dipole moment of p-dichlorobenzene is expected to be:

(A) 3.46 D (B) 0.00 D (C) 1.73 D (D) 1.00 D

CCB554

6. Which of the following would be expected to have a dipole moment of zero on the basis of symmetry?

(A) $SOCl_2$ (B) OF_2 (C) SeF_6 (D) ClF_5

CCB555

Types of Hydrogen Bonding

7. In which of the following compound, intra-molecular H-bonding is not observed :

(A) o-hydroxy benzyaldehyde (B) o-nitrophenol

(C) Chloral hydrate (D) Boric acid

CCB556

For More Join: @IITJEE_Advanced

8. How many of the following has hydrogen bonding

(a) NH_3 (b) CH_4 (c) H_2O (d) HI

(e) HF (f) HCOOH (g) $B(OH)_3$ (h) CH_3COOH

(i) HCO_3^- ion

CCB557

Applications of Hydrogen Bonding; Viscosity ; Boiling Point; Volatility; Solubility in Water

9. Amongst NH_3, PH_3, AsH_3 and SbH_3 the one with highest boiling point is :

(A) NH_3 because of lower molecular weight

(B) SbH_3 because of higher molecular weight

(C) PH_3 because of H-bonding

(D) AsH_3 because of lower molecular weight

CCB558

10. The correct order of boiling point is :

(A) $H_2O < H_2S < H_2Se < H_2Te$ (B) $H_2O > H_2Se > H_2Te > H_2S$

(C) $H_2O > H_2S > H_2Se > H_2Te$ (D) $H_2O > H_2Te > H_2Se > H_2S$

CCB559

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	84.50	(A) Q, S ; (B) P, Q, T ; (C)Q, S ; (D)Q, R, S	C	B	C	D	7	B	B

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Chemical Bonding : RACE-05

Inert Pair Effect

1. **Statement-1 :** Arsenic(V) chloride is relatively less stable than Phosphorus(V) chloride.

Statement-2 : Incomplete shielding of nucleus in Arsenic as compared to Phosphorus, thereby increasing the energy of 4s orbital in arsenic.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

CCB560

2. **Statement-1 :** Pb^{4+} compounds are stronger oxidizing agents than Sn^{4+} compounds.

Statement-2 : The higher oxidation states for the group 14 elements are more stable for the heavier members of the group due to 'inert pair effect.

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(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

CCB561

3. Which of the following occurs as a consequence of inert pair effect ?

(a) $SnCl_2$ act as a reducing agent (b) $SnCl_4$ act as an oxidizing agent

(c) SnO_2 is amphoteric (d) PbO_2 is an oxidant

(e) CCl_2 is unstable but $PbCl_2$ is stable

(A) a, d, e (B) d, e (C) a, b, c, d, e (D) a, b, c

CCB562

4. Find no. of ions which have ns^2np^0 configuration of outermost shell

Ga^+, Zn^{2+}, Ga^{3+}, Sn^{4+}, In^{3+}, Pb^{2+}, Tl^+

CCB563

Back Bonding

5. In which of the following compounds, the $p\pi-d\pi$ bonding take place?

(A) P_4O_{10} (B) HNO_3 (C) N_2O_5 (D) $HClO_3$

CCB564

For More Join: @IITJEE_Advanced

6. For BF_3 molecule which of the following is true ?

(A) B-atom is sp^2 hybridised.

(B) There is a $p\pi-p\pi$ back bonding in this molecule.

(C) Observed B–F bond length is found to be less than the expected bond length.

(D) All of these

CCB565

7. Which of the following statements regarding the structure of $SOCl_2$ is not correct?

(A) The Sulphur is sp^3 hybridised and it has a tetrahedral shape.

(B) The Sulphur is sp^3 hybridised and it has a trigonal pyramid shape.

(C) The Oxygen-Sulphur bond is $p\pi$-$d\pi$ bond.

(D) It contains one lone pair of electrons in the sp^3 hybrid orbital of sulphur.

CCB566

8. Respective order of strength of back-bonding and Lewis acidic strength in boron trihalides is:

(A) $BF_3 < BCl_3 < BBr_3$ and $BF_3 < BCl_3 < BBr_3$

(B) $BF_3 > BCl_3 > BBr_3$ and $BF_3 > BCl_3 > BBr_3$

(C) $BF_3 > BCl_3 > BBr_3$ and $BF_3 < BCl_3 < BBr_3$

(D) $BF_3 < BCl_3 < BBr_3$ and $BF_3 > BCl_3 > BBr_3$

CCB567

9. In which of the following compounds B–F bond length is shortest?

(A) BF_4^- (B) $BF_3 \leftarrow NH_3$ (C) BF_3 (D) $BF_3 \leftarrow N(CH_3)_3$

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CCB568

10. Which of the following is **not** due to back bonding?

(A) State of hybridisation may change (B) Bond order increases

(C) Bond angle always increases (D) Lewis acidity decreases

CCB569

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	C	B	3	AD	D	A	C	C	C

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Chemical Bonding : RACE-06

Bridge Bonding (3C-2e⁻, 3C-4e⁻), B_2H_6, Al_2Cl_6, Bond Behaviour of B_2H_6 ; Bridge Bonding in other Compounds; Reactions of Diborane (Symmetrical and Unsymmetrical cleavage, Methylation)

1. Electron deficient molecule among the following is:

(A) I_2Cl_6 (B) B_2H_6 (C) Al_2Cl_6 (D) All of these

CCB570

2. Which compounds are Lewis acids ?

(A) $AlCl_3$ (B) BCl_3 (C) H_2O (D) NH_3

CCB571

3. Which of the following is/are electron deficient compounds?

(A) $NaBH_4$ (B) B_2H_6 (C) $AlCl_3$ (D) $LiAlH_4$

CCB572

4. In which of the dimerisation process, the achievement of the octet is not the driving force.

(A) $2AlCl_3 \longrightarrow Al_2Cl_6$ (B) $BeCl_2 \longrightarrow BeCl_2$ (solid)

(C) $2ICl_3 \longrightarrow I_2Cl_6$ (D) $2NO_2 \longrightarrow N_2O_4$

CCB573

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5. The molecule is not having 3c – 2e bond.

(A) BeH_2 (dimer) (B) BeH_2 (solid) (C) C_2H_6 (D) B_2H_6

CCB574

Oxyacids of Phosphorus

6. Which of the following statements is **correct** regarding phosphoric acid ?

(A) pπ–dπ back bonding exist between O and P.

(B) The anion is resonance stablized.

(C) It is a dibasic acid.

(D) Inter molecular H bonding between molecules make it a syrupy (viscous) liquid.

CCB575

Oxyacids of Sulphur

7. **Match the following :**

	Column-I		**Column-II**
(A)	$H_3P_3O_9$	(p)	S–O–S bond is present
(B)	$H_2S_2O_7$	(q)	Di-basic acid
(C)	$H_2S_4O_6$	(r)	P–O–P bond is present
(D)	$H_4P_2O_5$	(s)	Central atom (S or P) in maximum oxidation state.

CCB576

For More Join: @IITJEE_Advanced

8. Difference in the oxidation number of sulphur atom is in $Na_2S_4O_6$ is x, that of $H_2S_2O_5$ is y. Find value of x × y is :

CCB577

9. Identify the correct statement

(A) $H_2S_2O_7$ has peroxy linkage

(B) $H_2S_2O_6$ has S–S linkage

(C) $H_2S_2O_8$ has peroxy linkage

(D) H_2SO_3(Sulphurous acid) has S in +4 oxidation state

CCB578

Oxyacids of Chlorine, Bromine, Iodine

10. **S_1 :** Oxidation number of N in N_2O_5 is 5

S_2 : The anhydride of Hypochlorous acid is Cl_2O

S_3 : As the electronegativity of central atom in a molecule having same hybridisation state and same terminal atoms increases, bond angle increases.

S_4 : For heteronuclear diatomic species A–B, the bond length decreases as the difference in electronegativity values increases (considering A and B of similar size) in all compound.

(A) T T T F (B) F T T T (C) F F T F (D) T T F T

CCB579

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	AB	BC	C	C	ABD	(A-r,s); (B–p,q,s) ; (C–q); (D–q,r)	10	BCD	D

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Chemical Bonding : RACE-07

Hydrolysis

1. In which of the following compounds B atoms are in sp^2 and sp^3 hybridisation states?
(A) Borax (B) Diborane (C) Borazole (D) All

CCB580

2. What is the change in hybridisation of Si when $SiCl_4$ form transition state during hydrolysis?
(A) $sp^3 \longrightarrow sp^3d$ (B) $sp^3d \longrightarrow sp^3$ (C) $sp^3 \longrightarrow sp^2$ (D) $sp^3 \longrightarrow sp^3$

CCB581

3. CCl_4 is inert toward hydrolysis under ordinary condition because :
(A) No vacant orbital on attacking site of C-atom
(B) H_2O molecule cannot approach to the vacant orbital of carbon due to steric crowding.
(C) Bond dissociation energy of C – Cl bond is very high
(D) CCl_4 is non-polar and does not react with polar H_2O molecule

CCB582

4. Select **INCORRECT** trend for hydrolysis.
(A) $SF_4 > SF_6$ (B) $NCl_3 > NF_3$ (C) $PF_3 > PCl_3$ (D) $SeF_6 > SF_6$

CCB583

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5. Total number of water molecule(s) consumed for complete hydrolysis of one molecule P_4O_{10} is :

CCB584

Introduction of Weak Forces; Types of Weak Forces; Factors Affecting Weak Forces; Comparison of Boiling Point of Species

6. Select the **CORRECT** statement for H_2 molecule
(A) On time average the molecule is non-polar but at the particular moment it may act as a dipole which is equally probable in all directions
(B) On time average the molecule is polar but at the particular moment it does not act as a dipole.
(C) On time average the molecule is non-polar and the particular moment it does not act as dipole.
(D) All are incorrect

CCB465

7. Select the **CORRECT** order of B.P.
(A) $BF_3 > BMe_3$ (B) $BF_3 < BMe_3$ (C) $BF_3 = BMe_3$ (D) None of these

CCB466

8. Select the **CORRECT** statement
(A) Boiling point of inert gases decreases down the group
(B) Boiling point of inert gases increases down the group
(C) Boiling point of H_2 < He
(D) None of these

CCB467

For More Join: @IITJEE_Advanced

9. **Statement-1** : Experimentally 100 % covalent bond formation is not possible

Statement-2 : Non-polar molecule has instantaneous dipole – induced dipole interaction

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CCB468

10. London force works in

(A) Polar molecule

(B) Non-polar molecule

(C) All polar and non-polar molecule

(D) Only in polar molecule

CCB470

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	A	A	C	6	A	B	B	A	ABC

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Chemical Bonding : RACE-08

Molecular Orbital Theory; Drawbacks of Valence Bond Theory; Introduction of Molecular Orbital Theory

1. If N_B is the number of bonding electrons and N_A is the number of antibonding electrons of a molecule. Then choose the **incorrect** statement(s) for the given relationship, $N_B > N_A$

(A) Molecule must be diamagnetic

(B) Molecule may have any integral, fractional or zero value of bond order

(C) Molecule is only paramagnetic species

(D) Molecule does not exist

CCB481

Linear Combination of Atomic Orbitals

2. How many of the following are gerade molecular orbitals?

(i) σ^*2s (ii) $\sigma 2p_z$ (iii) $\pi 2p_y$ (iv) π^*2p_x

CCB585

Bonding in some homonuclear diatomic molecules (Molecular Orbital Diagram of H_2, He_2, Li_2, O_2, B_2, C_2, N_2, F_2, Ne_2)

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3. Which of the following molecules/ions exhibit sp mixing?

(A) B_2 (B) C_2^{2-} (C) O_2^+ (D) Both (A) and (B)

CCB587

4. The species which are diamagnetic :

(A) O_2^- (B) NO_2 (C) ClO_2 (D) N_2O_4

CCB588

Applications of MOT (Molecular Behaviour) ; 1. Bond Order; 2. Bond Length; 3. Bond Strength; 4. Ionisation Energy; 5. Colour of Halogen Molecules

5. Select the **CORRECT** order -

(A) Bond length = $O_2^{2-} > O_2^- > O_2 > O_2^+$

(B) Bond strength = $O_2^+ > O_2 > O_2^- > O_2^{2-}$

(C) Unpaired electron(s) = $O_2 > O_2^+ = O_2^- > O_2^{2-}$

(D) No. of antibonding electron(s) = $O_2^{2-} > O_2^- > O_2 > O_2^+$

CCB482

6. What is the bond order of underlined species in $\underline{NO}$ $[BF_4]$?

CCB589

7. Which of the following pairs have identical values of bond order?

(A) N_2^+ and O_2^+ (B) F_2 and Ne_2 (C) O_2 and B_2 (D) C_2 and N_2

CCB590

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8. Among the following species, which has the minimum bond length ?
(A) B_2 (B) C_2 (C) F_2 (D) O_2^-

CCB591

Concept of Hydration; Hydration Energy; Hydrated Radius; Ionic Mobility in Water; Conductivity in Water; Condition for Solubility of Ionic Compounds in Water

9. Lattice enthalpy and enthalpy of solution of NaCl are 788 kJ mol^{-1} and 4 kJ mol^{-1}, respectively. The hydration enthalpy of NaCl is (magnitude only) :

CCB592

10. Atomic radii of alkali metals (M) follow the order Li < Na < K < Rb but ionic radii in aqueous solution follow the reverse order $Li^+ > Na^+ > K^+ > Rb^+$. The reason of the reverse order :
(A) increases in the ionisation energy
(B) decreases in the metallic bond character
(C) increases in the electropositive character
(D) decreases in the amount of hydration

CCB593

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	ABCD	2	D	D	ABCD	3	A	B	784	D

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RACE
Regular Analysis Through Continuous Exercise

Chemical Bonding : RACE-09

Fajan's Rule; Polarisation; Polarising Power; Polarizability; Factors Affecting Polarisation

1. Identify the incorrect order.

(A) $Zn^{+2} < Cd^{+2} < Hg^{+2}$ (ionic radius)

(B) $Zn^{+2} < Cd^{+2} < Hg^{+2}$ (polarizing power)

(C) $Zn^{+2} < Cd^{+2} < Hg^{+2}$ (Effective nuclear charge)

(D) $Zn^{+2} \neq Cd^{+2} \neq Hg^{+2}$ (Number of 4s e^-s)

CCB479

2. Find number of ions which have pseudo inert gas configuration.

Zn^{2+}, Ga^{3+}, Ge^{2+}, Ge^{4+}, Cu^{2+}

CCB484

3. Which of the following combination of ion will have highest polarisation?

(A) Fe^{2+}, Br^- (B) Ni^{4+}, Br^- (C) Ni^{2+}, Br^- (D) Fe, Br^-

CCB595

4. The correct order of decreasing polarizability of ion is :

(A) Cl^-, Br^-, I^-, F^- (B) F^-, I^-, Br^-, Cl^- (C) I^-, Br^-, Cl^-, F^- (D) F^-, Cl^-, Br^-, I^-

CCB596

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Properties Affected by Polarisation; Covalent Character of Ionic Compounds

5. Out of the following which one has the highest values of covalent character

(A) $ZnCl_2$ (B) $CdCl_2$ (C) $HgCl_2$ (D) CuCl

CCB475

6. Find the number of compounds which are more covalent in nature compared to $AlCl_3$ from the following.

$AlBr_3$. AlI_3 . $BeCl_2$. $BeBr_2$. AlF_3 . BeI_2 [Given : $r_{Be^{2+}} : 0.31\,Å$, $r_{Al^{3+}} : 0.5\,Å$]

CCB483

7. Which of the following is in order of increasing covalent character?

(A) $CCl_4 < BeCl_2 < BCl_3 < LiCl$ (B) $LiCl < CCl_4 < BeCl_2 < BCl_3$

(C) $LiCl < BeCl_2 < BCl_3 < CCl_4$ (D) $LiCl < BeCl_2 < CCl_4 < BCl_3$

CCB597

Properties Affected by Polarisation; Melting Point of Ionic Compounds

8. Compound having lowest Melting point.

(A) $BeCl_2$ (B) $MgCl_2$ (C) $CaCl_2$ (D) $SrCl_2$

CCB476

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Properties Affected by Polarisation; Solubility of Ionic Compounds in Water

9. Which of the following order is correct of the given property.

(A) LiCl >NaCl > KCl > RbCl > CsCl : Thermal stability order

(B) $BeF_2 < MgF_2 < CaF_2 < SrF_2 < BaF_2$: solubility order

(C) $NO_2 > N_2O_5 > NO > N_2O$: acidic order

(D) BaO > SrO > CaO > BeO > MgO : basic character order

CCB477

10. Identify the correct order.

(A) CsCl < RbCl < KCl < NaCl < LiCl (Solubility in water)

(B) CsCl < RbCl < KCl < NaCl < LiCl (m.p)

(C) CsCl > RbCl > KCl > NaCl > LiCl (% ionic character)

(D) CsCl > RbCl > KCl > NaCl > LiCl (Lattice Energy)

CCB478

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	3	B	C	C	5	C	A	A	C

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